

Bhavdeep Arora

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SUMMARY

Computer Science graduate specializing in Software Engineering with a track record of delivering data-driven projects across ML, cybersecurity, and climate tech. Known for strong communication, collaborative problem-solving, and the ability to ship clean, well-documented work independently.

EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science (Honours) Computer Science · Software Engineering Concentration

2022–2026

- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Data Science

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, HTML/CSS, Go, Rust

Frameworks: React, Node.js, Tailwind, Django, Flask

Developer Tools: Git, Docker, Google Cloud Platform, TravisCI

Data & Cloud: Hadoop, Apache Spark, MySQL, MongoDB, SQLite, AWS, Azure, Firebase

PROJECTS

Lacunae | *Python, PyTorch, U-Net, fastMRI, HDF5* | [Github](#)

- Trained a **U-Net** (7.7M parameters) on the fastMRI single-coil knee dataset to reconstruct diagnostically useful MRI images from **4x undersampled k-space data**, recovering structure that naive inverse FFT cannot
- Implemented a **random Cartesian undersampling mask** with fixed low-frequency center retention, trained with L1 loss and evaluated using **SSIM/PSNR metrics** against fully sampled ground truth

Deadzone | *Python, Random Forest, ERA5/NOAA Datasets, Plotly Dash* | [Github](#)

- Engineered an end-to-end **ML pipeline** analyzing Gulf of Mexico hypoxic dead zones using ERA5 and NOAA datasets, training a **Random Forest model** to predict dead zone extent from climate and nutrient runoff features
- Built an **interactive Plotly Dash dashboard** to visualize spatial dead zone trends and model outputs across multi-decade environmental data

Thunderhead | *Go, Reverse Proxy, HTTP, Behavioral Analysis* | [Github](#)

- Built a **lightweight reverse proxy** in Go that scores incoming HTTP requests 0–100 across five behavioral signals (robots.txt violations, crawl patterns, request rate, header anomalies) to detect and mitigate bot traffic without third-party services
- Implemented **graduated response dispatch** — allow, tarpit (5s delay), or 403 block — based on configurable score thresholds, with structured JSON logging of all decisions

Terraseed | *Python, Machine Learning, Climate Datasets, Plotly Dash* | [Github](#)

- Developed a **global planting window predictor** using ML classification models trained on large-scale gridded climate data to generate region-specific crop recommendations by temperature, precipitation, and frost risk
- Delivered results through an **interactive geospatial dashboard** allowing users to query optimal planting windows by location and crop type

EXPERIENCE & LEADERSHIP

Inspire | *Speaker Coordinator Team Leader*

January 2023 - March 2023

- Led a team of coordinators to source and onboard speakers for a university-scale event, managing outreach and scheduling end-to-end

Sci Exchange at Toronto Metropolitan University | *Volunteer & Assistant*

September 2023 - April 2024

- Mentored and guided 50+ participants through science exchange exhibits, translating complex topics into accessible explanations